

G.E.T.S. Future vs. Future

Arbitrage Opportunity Model

User Manual

Version 1.0.0.1

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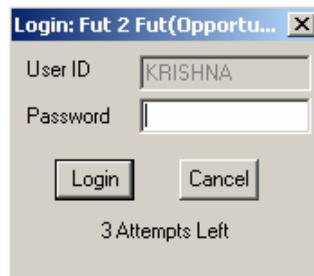
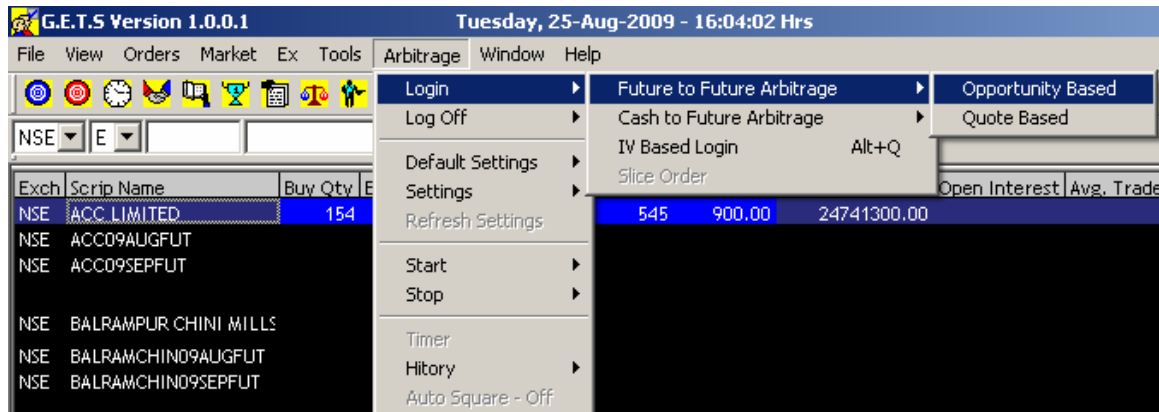
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Future vs. Future arbitrage Opportunity model is used for rollover of future positions across different expiries for any derivative counter. The advantage of opportunity model is to catch the accurate spread expected or above and faster execution in capturing the opportunity.

I. Login:

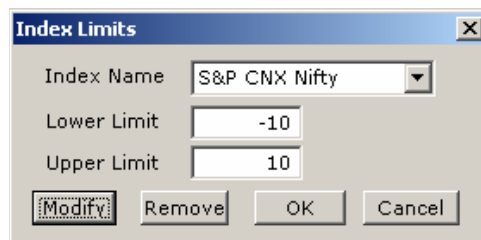
The user needs to login through GETS CTCL client, after which he will get the access of future vs. future opportunity model from arbitrage menu as shown below.



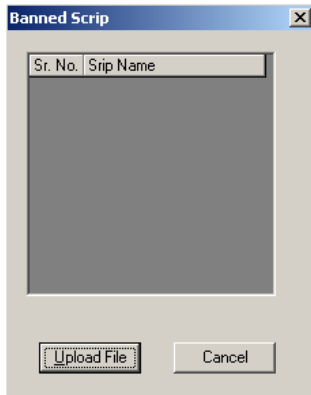
If the user enters wrong password for more than 3 times, his login will be blocked. To unblock, he has to request the system administrator.

During the login sequence to Future vs. future arbitrage opportunity model, following checks must be defined, which will be validated in the post-login and execution.

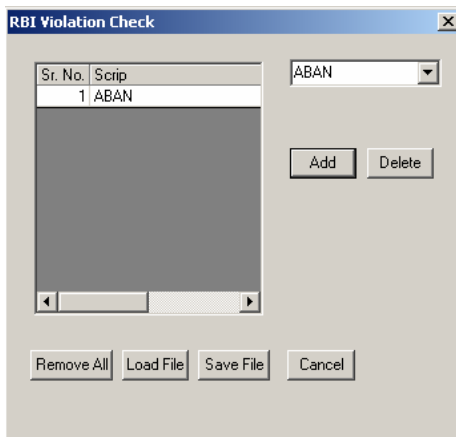
Index Limits Check – When index crosses the set volatility range, then automated arbitrage will be stopped. Need to define the index and volatility range.



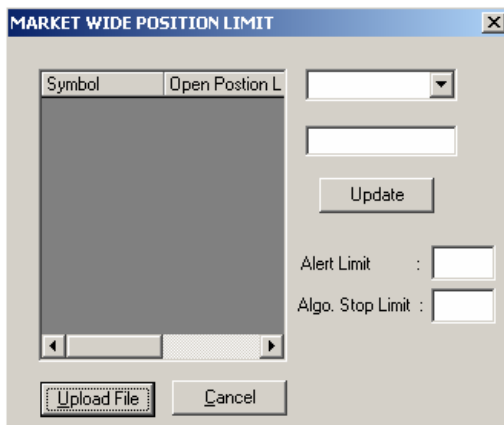
Banned Scrip Check – Trading in any banned scrip can be ruled out by uploading the banned scrip list. All violated symbols won't be loaded in the arbitrage settings form.



RBI Violation Check – Trading in any RBI violated scrip can be ruled out by uploading RBI violation list. All violated symbols won't be loaded in the arbitrage settings form.



Market Wide Position Limit Check – Trading in contracts, where market wide position limit reached, will be ruled out by uploading the market wide position limit list. All violated symbols won't be loaded in the arbitrage settings form.



II. Input parameters:

After successful login, the user will be prompted with the Rollover limit settings dialog as shown below. He can call the dialog from market watch using short key CTRL+F9.

Symbol	ProC	Account Id	Buy Expiry	Sell Expiry	Lower Limit	Ord Qty	Total Qty	Traded Qty	Rem Qty	Lotsize
NIFTY	PRO	07782	24SEP2009	27AUG2009	-5.00	10	50	0	2500	50
NIFTY	PRO	07782	27AUG2009	24SEP2009	10.00	10	50	0	2500	50

All input parameters are explained below.

1. Exchange: NSE
2. Symbol: Derivative symbols listed on NSE
3. PRO/CLI: Choose for proprietary trading / client trading
4. Account ID: Select the client account ID
5. Participant ID: NSE participant ID, in case of institutional client
6. Near Expiry: Derivative near expiry date
7. Far Expiry: Derivative far expiry date
8. Lot Size: Lot size of the symbol will be selected automatically

Settings for Buy Near & Sell Far (Short Roll): Check Buy Near :: Sell Far

9. Lower Limit: Input required absolute spread range
10. Order Lots: Input the no. of lots that has to be placed for each order
11. Total Lots: Input the total no. of lots to execute
12. Traded Qty: Displays the traded quantity for selected symbol
13. Remaining Qty: Displays the remaining quantity for selected symbol

Settings for Buy Far & Sell Near (Long Roll): Check Buy Far :: Sell Near

14. Lower Limit: Input required absolute spread range
15. Order Lots: Input the no. of lots that has to be placed for each order
16. Total Lots: Input the total no. of lots to execute
17. Traded Qty: Displays the traded quantity for selected symbol
18. Remaining Qty: Displays the remaining quantity for selected symbol

III. Controlling buttons:



Add: To add the new arbitrage opportunity setting to the arbitrage farm

Modify: To modify any previously added settings to the arbitrage farm

Remove: To remove any previously added settings from the arbitrage farm

Remove All: Removes all arbitrage settings from the arbitrage farm

Cancel: To cancel any partially filled arbitrage setting

Save File: To save all the arbitrage settings present in the arbitrage farm

Load File: To load previously saved arbitrage settings to arbitrage farm

Start / Stop: To start calculating the opportunity and submission of the orders / to stop calculating the opportunity. After calling stop button, all pending orders from order book operated by the arbitrage module will be canceled.

At any point of time the user can add maximum of 20 symbols and trading happens simultaneously across all counters.

IV. Execution Logic:

After filling the input parameters, the user must call the Start control button. This activates the application to calculate the live spread in market and compares with the set price difference. If the live spread is equal to or greater than the set spread, then a 2LEG order with order validity as IOC having two legs (first leg as far expiry and second leg as near expiry) will be placed for the available lots in the market.

The calculation of live spread is based on below formulae.

Buy Near :: Sell Far

Calculated live spread = (Best Bid of Far expiry – Best Ask of Near expiry)

Sell Near :: Buy Far

Calculated live spread = (Best Bid of Near expiry – Best Ask of Far expiry)

In case of multiple order lots, weighted average price is considered based on the market depth.

As the order validity is IOC, either both the legs gets executed or entire order gets cancelled. So there will be no risk of partial execution on any side, thus ruling out the price risk.

V. Order Book & Trade Book:

All the orders resulting from the execution of future vs. future arbitrage opportunity model gets updated in pending order book (F3). And all trades get updated in previous trade book (F8).

Pending Orders Of All Scripts																				
GreekC	RefNo.	Exch	Trader	Inst	Na	Code	Symb	Ser/Exp/Grc	Strik	Op	B/S	Pend	Price	Total	Prc	Clie	Client N.	Exch. Order No.	Settlor	Valid
AJAY1	200002808	NSE	18192	F	FUTIDX	8986	NIFTY	27AUG2009			Buy	50	0.00	50	CLI	0H01	HITESH	2009080310000122	07782	DAY
AJAY1	200002808	NSE	18192	F	FUTIDX	5999	NIFTY	24SEP2009			Sell	50	0.00	50	CLI	0H01	HITESH	2009080310000122	07782	DAY
AJAY1	200002807	NSE	18192	F	FUTIDX	8986	NIFTY	27AUG2009			Buy	50	0.00	50	CLI	0H01	HITESH	2009080310000121	07782	DAY
AJAY1	200002807	NSE	18192	F	FUTIDX	5999	NIFTY	24SEP2009			Sell	50	0.00	50	CLI	0H01	HITESH	2009080310000121	07782	DAY
AJAY1	200002806	NSE	18192	F	FUTIDX	8986	NIFTY	27AUG2009			Buy	50	0.00	50	CLI	0H01	HITESH	2009080310000120	07782	DAY
AJAY1	200002806	NSE	18192	F	FUTIDX	5999	NIFTY	24SEP2009			Sell	50	0.00	50	CLI	0H01	HITESH	2009080310000120	07782	DAY
AJAY1	200002811	NSE	18192	F	FUTIDX	8986	NIFTY	27AUG2009			Buy	50	0.00	50	CLI	0H01	HITESH	2009080310000125	07782	DAY
AJAY1	200002811	NSE	18192	F	FUTIDX	5999	NIFTY	24SEP2009			Sell	50	0.00	50	CLI	0H01	HITESH	2009080310000125	07782	DAY
AJAY1	200002810	NSE	18192	F	FUTIDX	8986	NIFTY	27AUG2009			Buy	50	0.00	50	CLI	0H01	HITESH	2009080310000124	07782	DAY
AJAY1	200002810	NSE	18192	F	FUTIDX	5999	NIFTY	24SEP2009			Sell	50	0.00	50	CLI	0H01	HITESH	2009080310000124	07782	DAY
G.E.T.S Pending Orders Of All Scripts NIFTY09AUGFUT												BuyQty : 3100			SellQty : 0					

Trades-All																	
Exch	Membr	Trader	Inst	Na	Symbol/Sci	Ser/Exp/Grc	Strike	Opt	B/S	Qty	Price	Pro	Exch. Order No	Trader	Time	GreekC	Ref No
NSE	07782	18192	F	FUTIDX	NIFTY	24SEP2009			Sell	50	4691.70	CLI	09080410000044	20019	12:13:50	AJAY1	100000071
NSE	07782	18192	F	FUTIDX	NIFTY	24SEP2009			Sell	50	4691.70	CLI	09080410000044	20018	12:13:50	AJAY1	100000071
NSE	07782	18192	F	FUTIDX	NIFTY	24SEP2009			Sell	50	4691.70	CLI	09080410000044	20017	12:13:50	AJAY1	100000071
NSE	07782	18192	F	FUTIDX	NIFTY	27AUG2009			Buy	50	4689.00	CLI	09080410000040	20016	12:13:44	AJAY1	100000067
NSE	07782	18192	F	FUTIDX	NIFTY	24SEP2009			Sell	50	4692.35	CLI	09080410000042	20015	12:13:42	AJAY1	100000069
NSE	07782	18192	F	FUTIDX	NIFTY	27AUG2009			Buy	50	4689.25	CLI	09080410000041	20014	12:13:42	AJAY1	100000068
NSE	07782	18192	O	OPTIDX	NIFTY	27AUG2009	4600.00	PE	Sell	50	128.30	CLI	09080410000039	20013	12:13:18	AJAY1	100000066
NSE	07782	18192	O	OPTIDX	NIFTY	27AUG2009	4600.00	CE	Buy	50	214.90	CLI	09080410000038	20012	12:13:16	AJAY1	100000065
NSE	07782	18192	F	FUTSTK	RELIANCE	27AUG2009			Buy	150	2043.70	CLI	09080410000037	20010	12:10:55	AJAY1	100000064
NSE	07782	18192	F	FUTSTK	RELIANCE	27AUG2009			Sell	150	2045.85	CLI	09080410000032	20004	12:10:14	AJAY1	100000059
NSE	07782	18192	F	FUTSTK	RELIANCE	27AUG2009			Buy	150	2046.00	CLI	09080410000031	20003	12:10:04	AJAY1	100000058

VI. Net Positions:

All the position summary of previous trades updated in net position dialog (ALT+F6)

Net Position																		
Exchange	Mkt Type	Symbol		Periodicity	User	Security / Contract												
NSE	All	All		Daily	AJAY1	Refresh												
Excl	User	Client	Mt	Instrument	Symbol	Ser/Exp	Strike	P	Op	Buy Qt	Buy Val	Buy Avg	Sell Qt	Sell Val	Sell Avg	Net	Net Val	MTM G/L
NSE	AJAY1	07782	O	OPTIDX	NIFTY	27AUG2009	4600.00	CE		50	10745.00	214.90	0	0.00	0.00	50	10745.00	357.50
NSE	AJAY1	07782	O	OPTIDX	NIFTY	27AUG2009	4600.00	PE		0	0.00	0.00	50	6415.00	128.30	-50	-6415.00	237.50
NSE	AJAY1	07782	F	FUTIDX	NIFTY	27AUG2009				100	468912.50	4689.13	0	0.00	0.00	100	468912.50	1257.00
NSE	AJAY1	07782	F	FUTIDX	NIFTY	24SEP2009				0	0.00	0.00	550	2580467.50	4691.76	-550	2580467.50	-7282.00
NSE	AJAY1	07782	F	FUTSTK	RELIANCE	27AUG2009				300	613455.00	2044.85	150	306877.50	2045.85	150	306577.50	532.50
Tota										450	093112.50		750	2893760.00		-300	1800647.50	-4897.50
G.E.T.S	Net Position		NIFTY09AUG4600CE			Buy Qty: 50		Sell Qty: 0		Net Qty: 50		Buy Val: 10745.00		Sell Val: 0.00		Net Val: 10745.00 MTM: 405.00		

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